

Mohak Kapoor

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SUMMARY

ML Engineer with 1+ year shipping production ML systems across edge, serverless, and containerized environments. I build end-to-end data pipelines and multi-model systems from ingestion to Dockerized deployment on real hardware. Built a two-stage hybrid IDS on 2.8M rows where LightGBM hit 99% accuracy (F1: 0.99) in a live FastAPI service.

PROJECTS

METR — Market Exposure Timing Research — Python, Polars, SQL, XGBoost, Feature Engineering *Mar-Apr 2026*

- Constructed a **two-layer meta-labeling framework** on 3,440+ days of multi-asset Indian market data (Nifty 50, GoldBees, USD/INR), using Polars and **SQL-based validation queries** with signal-conditional masking and 30 cross-asset features for **time-series classification** and trade entry quality filtering via XGBoost.
- Validated against **10,000 Monte Carlo simulations**. GoldBees filter achieved **Sharpe 1.48, Calmar 1.87, 66.1% win rate OOS (2024–2025)**, significant at $p < 0.05$ across three independent tests; null results on Nifty and USD/INR consistent with market efficiency theory.
- Applied **Fractional Differentiation** ($d=0.45$) preserving 81% historical memory with stationarity, **Triple Barrier volatility-adaptive labeling**, Exploratory Data Analysis and **SHAP feature importance** analysis identifying Vol Efficiency and VIX Momentum as primary alpha drivers consistent with GoldBees' rupee-dollar exposure.

Network IDS Detection Engine — Python, LightGBM, XGBoost, PyTorch, scikit-learn, FastAPI, Docker *Jan-May 2026*

- Engineered a **two-stage hybrid IDS** on 2.8M rows of CICIDS2017. Unsupervised anomaly detection flags candidates for supervised classification, mirroring production platforms like Snort and Darktrace; built **end-to-end data pipeline** handling ingestion, EDA, preprocessing, and temporal train/test splitting to prevent distribution leakage across unseen attack types.
- Benchmarked LightGBM at **99% accuracy (F1: 0.99)**, XGBoost 97%, PyTorch FFNN 98%, SVM 96.7%. Denoising Autoencoder (AUC: 0.7801) outperforms Isolation Forest (AUC: 0.7156) on zero-day detection; IncrementalPCA **reduced feature space 51%** (69→34 features, 98.97% variance retained) across dual preprocessing paths.
- Shipped **Dockerized FastAPI inference engine** on Raspberry Pi 5 with **real-time WebSocket streaming dashboard** for live model deployment with predictions, latency metrics, and threat statistics streamed to deployed web UI; exposed publicly via **Cloudflare Tunnel** with Bearer token auth.

CaptchaOCR — CAPTCHA Recognition System — Python, PyTorch, CNN, BiLSTM, Computer Vision *Dec 2025*

- Trained a CAPTCHA recognition system using a **CNN + BiLSTM (CRNN) architecture with CTC loss**, achieving **96%+ character-level accuracy** on distorted, variable-length sequences via synthetic training corpus generation and noise-augmentation preprocessing.
- Published to **Hugging Face Spaces** for public **model deployment** and inference, enabling reproducible benchmarking and open community validation beyond offline experimentation.

EXPERIENCE

HumanizeIQ *June 2025 – October 2025*
AI Intern *Hazlet, New Jersey (Remote)*

- Engineered production **MCP servers** on **Cloudflare Workers** for client-facing async image generation and multimodal document workflows, consolidating **3 manual pipeline steps** into a single API call via persistent R2 storage.
- Built **LLM pipelines (GPT-4, Gemini)** for automated call transcription and structured report generation, eliminating manual post-call processing and **cutting analyst turnaround from ~2 hours to under 5 minutes**.
- Designed **REST APIs** for async job orchestration enabling product teams to run long-running LLM inference reliably within chat workflows, supporting **100% of the platform's background processing load**.

TECHNICAL SKILLS

Languages	Python (Advanced), TypeScript, C++
ML & Modeling	Supervised Learning, SVMs, LightGBM, XGBoost, CNN, BiLSTM, CRNN, Time-Series Forecasting, Anomaly Detection, Statistical Modeling, Feature Engineering, Cross-Validation, Hyperparameter Tuning
LLMs & GenAI	LLM Integration (GPT-4, Gemini), Multimodal Pipelines, MCP, RAG
MLOps & Deployment	FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), MLflow, Experiment Tracking, Model Monitoring, Cloudflare Workers, Cloudflare Tunnel, Edge Deployment (Raspberry Pi 5)
Frameworks & Data	TensorFlow, PyTorch, scikit-learn, Django, Hugging Face, Pandas, NumPy, Polars, PostgreSQL, SQL, Jupyter, Git, Linux
CS Fundamentals	Data Structures & Algorithms, Object-Oriented Programming, Computer Networks

EDUCATION

Jaypee Institute of Information Technology *Noida, India*
B.Tech in Electronics and Communication Engineering *2022-2026*

- Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Computer Systems, Multivariate & Differential Calculus, Control Systems